



ACADEMIC PROJECT · 2026

# Requirements Specification

NBP Currency Exchange Rates  
Currency Analytics System (CAS)

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## Product Owner

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## Project Team

ZPI2025\_IO2\_Wladca\_Requestow\_Druzyna\_Commita

- **Tymoteusz Kot** - SCRUM Master / Developer
- **Urszula Szmit** - DevOps
- **Szymon Pokora** - Developer
- **Tobiasz Grala** - Tester

# 1. Introduction

## Purpose of This Document

The purpose of this document is to define the software requirements for the **Currency Analytics System (CAS)**, a command-line application designed to perform statistical analysis and calculations on currency exchange rates.

The system will retrieve historical and real-time data strictly from the official National Bank of Poland (NBP) API accessible at <https://api.nbp.pl/>

This document outlines both the **functional capabilities** the system must possess and the **non-functional constraints** under which it must operate.

## 1.1. Project Context

### Methodology

- 1 The project is developed by a four-person academic team utilizing the **SCRUM methodology**.

### Interface

- 2 Because the team consists of fewer than five members, a GUI is **not mandatory**; communication with the user will be handled via a **command-line interface (CLI)**.

### Language

- 3 All documentation, including this specification, GitHub commits, backlog items, and code comments, is written entirely in **English**.



## 2. Functional Requirements

The functional requirements describe the specific behaviors, calculations, and interactions the system must support.

### 2.1. Data Acquisition (NBP API Integration)

- The system shall fetch all required currency exchange data directly from the NBP Web API (<https://api.nbp.pl/>)
- The set of currencies available for analysis is strictly limited to the major foreign currencies published in the NBP's standard Table A of average exchange rates (e.g., EUR, USD, CHF, GBP).
- The system must parse JSON or XML responses from the NBP API.
- The system shall handle network timeouts, invalid responses, and API rate limits gracefully, displaying appropriate error messages to the user without crashing.
- The system must support querying data for specific timeframes matching the required analysis periods.

### 2.2. User Interface & Interaction

- Communication between the user and the system shall take place via a console (CLI).
- The system must allow the user to specify detailed parameters for the statistical analysis.
- The user must be able to input the type of analysis desired (Session Analysis, Statistical Measures, or Distribution Analysis).
- The user must be able to input the settlement period for the analysis.
- The user must be able to define the specific currency or currency pairs for the calculation.

### 2.3. Session Analysis Module

- The system must calculate and display the total number of rising sessions, falling sessions, and unchanged sessions.
- The analysis entails calculating the changes in consecutive sessions compared to the previous sessions.
- The system must sum these session outcomes within the calculated period.
- The system shall perform this session analysis for the following user-selected periods: the last 1 week, 2 weeks, 1 month, 1 quarter, half a year, and 1 year. This calculation must be performed for a single currency chosen by the user.

## 2.4. Statistical Measures Module

- The system must calculate specific statistical measures for the exchange rates. The required measures include the median and the mode.
- The system must calculate the standard deviation, utilizing the sample standard deviation formula.
- The system must calculate the coefficient of variation.
- These statistical measures must be available for the following periods: the last 1 week, 2 weeks, 1 month, 1 quarter, half a year, and 1 year.
- These measures must be calculated for a single currency selected by the user.

## 2.5. Distribution of Changes Module

- The system must calculate the distribution of changes for chosen currency pairs (e.g., EUR/USD).
- The analysis is restricted to monthly and quarterly changes.
- The system must represent the frequency of value changes within specified intervals as a histogram drawn in the console using ASCII art (e.g., using '\*' characters) to resemble a pyramid structure
- The distribution data output must clearly show the calculated ranges and the corresponding count of occurrences within each range. A "range" is defined as a mathematically calculated sub-interval of the total difference between the minimum and maximum currency exchange rate changes during the specified period.
- The entire span of changes will be divided into a fixed number of 13 equal intervals (ranges) to accurately reflect the data distribution and match the expected visual pyramid layout.

## 2.6. Output and Export Formats

- The system must be capable of presenting the results of the analyses conducted in various forms.
- The system should print the results directly to the console in a structured table format.
- The system shall allow the user to export the output data into a CSV (Comma-Separated Values) file format for external spreadsheet analysis.



# 3. Non-Functional Requirements

The non-functional requirements define system attributes such as architecture, development processes, testing, and deployment.

## 3.1. Methodology and Management

- The project workflow must involve at least two distinct Sprints, including at least one Sprint Review conducted entirely in English with the Product Owner's participation.
- All issues, sprint reports, comments, and project documentation must be written in English.

## 3.2. Version Control System (VCS)

- The source code must be hosted in a repository on the GitHub platform. The team is strictly obligated to use a source code version control system.
- The repository branches must adhere to the GitFlow branching model, containing at minimum main, release, and develop branches.
- New project releases must be tagged starting from version 2.0.0.
- Every pull request merged into the release branch must automatically generate a new tagged release (incremented from 2.0.0).

## 3.3. Continuous Integration and Deployment (CI/CD)

- The team must configure a Continuous Integration (CI) system pipeline.
- The CI pipeline must strictly run only for the main, release, and develop branches.
- The instructor (Product Owner) must be granted full access to the configured CI system.
- The SCRUM Master must include the CI system configuration details within the repository's readme.md file.
- A successful merger to the release branch (assuming unit and integration tests pass) must result in a deployment or the generation of an executable file.



## 3.4. Testing and Quality Assurance

- Unit tests, integration tests (verifying the correct connection and data retrieval from the external NBP API), and functional tests must be implemented for the core system logic.
- The CI pipeline must execute the unit tests automatically.
- The team must provide at least one formal report from the acceptance tests carried out.
- The testing report must include documentation on bug detection, bug resolution, and a detailed description of the integration tests performed.
- The code must be readable, transparent, and appropriately commented in English.

## 3.5. Documentation and Repository Structure

- The repository must contain the application's source codes. The repository must contain a comprehensive readme.md file outlining the technology stack, deployment instructions, documentation location, backlog location, CI automation details, and test report locations.
- The repository must contain a dedicated folder for all project documentation. This documentation folder must include a detailed User Manual (instruction guide) explaining how to operate the CLI and interpret the results.
- The project documentation must include system architecture descriptions, component diagrams, sequence diagrams, and activity diagrams.

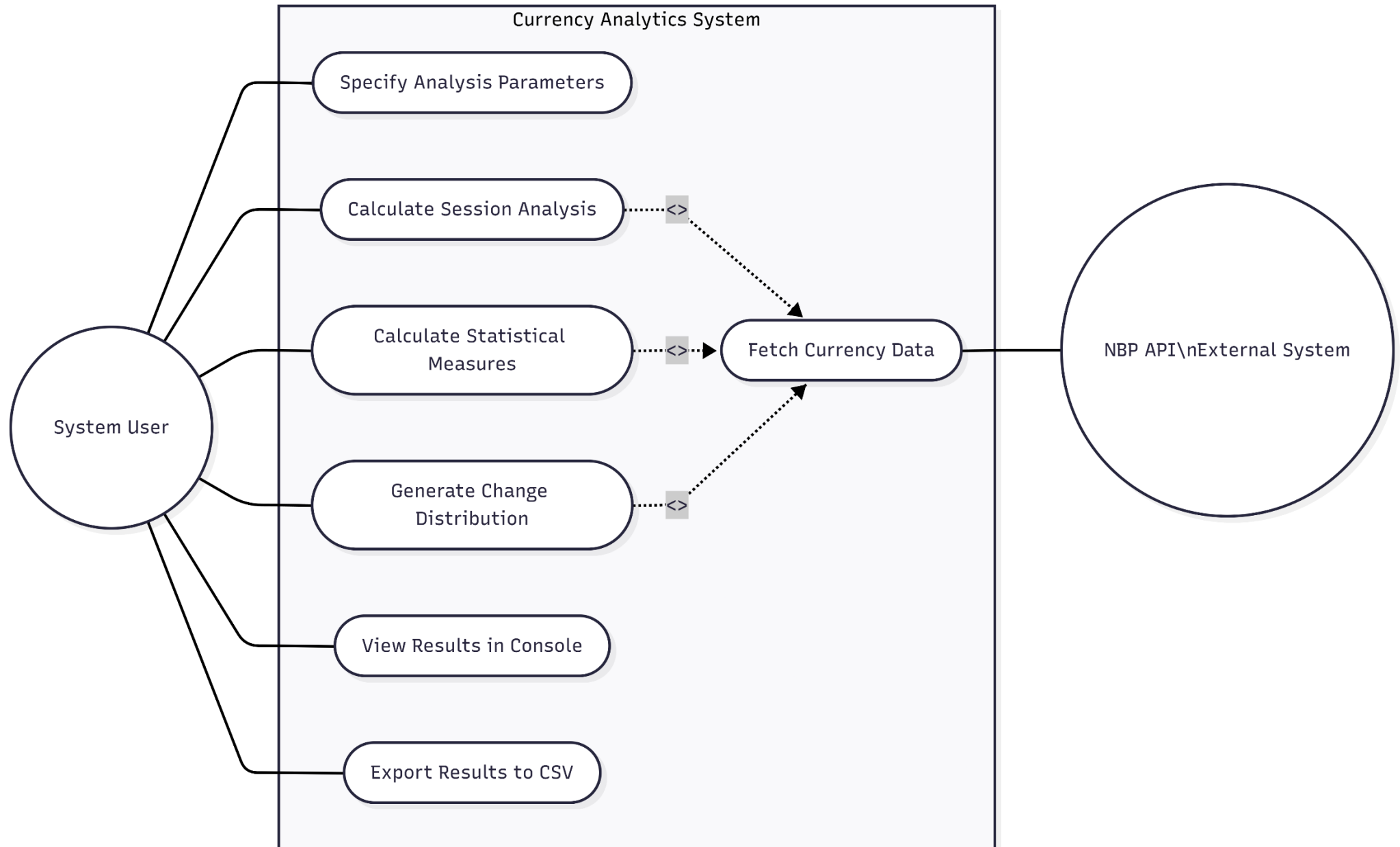


## 3.6. System Constraints / Attributes

- **Reliability:** The system must handle NBP API unresponsiveness, rate limits, or lack of internet connection gracefully by displaying a user-friendly error message in the console, rather than crashing unexpectedly.
- **Data Accuracy:** All calculated statistical measures and currency exchange rate changes must be rounded and displayed to at least four decimal places to ensure financial precision and readability.
- **Usability:** (CLI) must provide intuitive, step-by-step prompts for user inputs. In case of invalid input, the system must immediately reject it and display a clear, corrective hint rather than terminating the process.
- **Maintainability:** The system's source code must follow modular architecture, strictly separating the NBP API data fetching logic, statistical calculations, and CLI rendering. This ensures that future additions of new statistical measures or API endpoints can be integrated with minimal impact on existing code.
- **Third-Party Dependency (API Limitations):** The system's core functionality is strictly dependent on the availability of the external NBP API. Consequently, the system is subject to the API's inherent constraints, including a strict limit on historical data retrieval (a maximum of 367 days per single request), rate limiting, and potential temporary data unavailability caused by external server maintenance or 3rd-party downtimes.

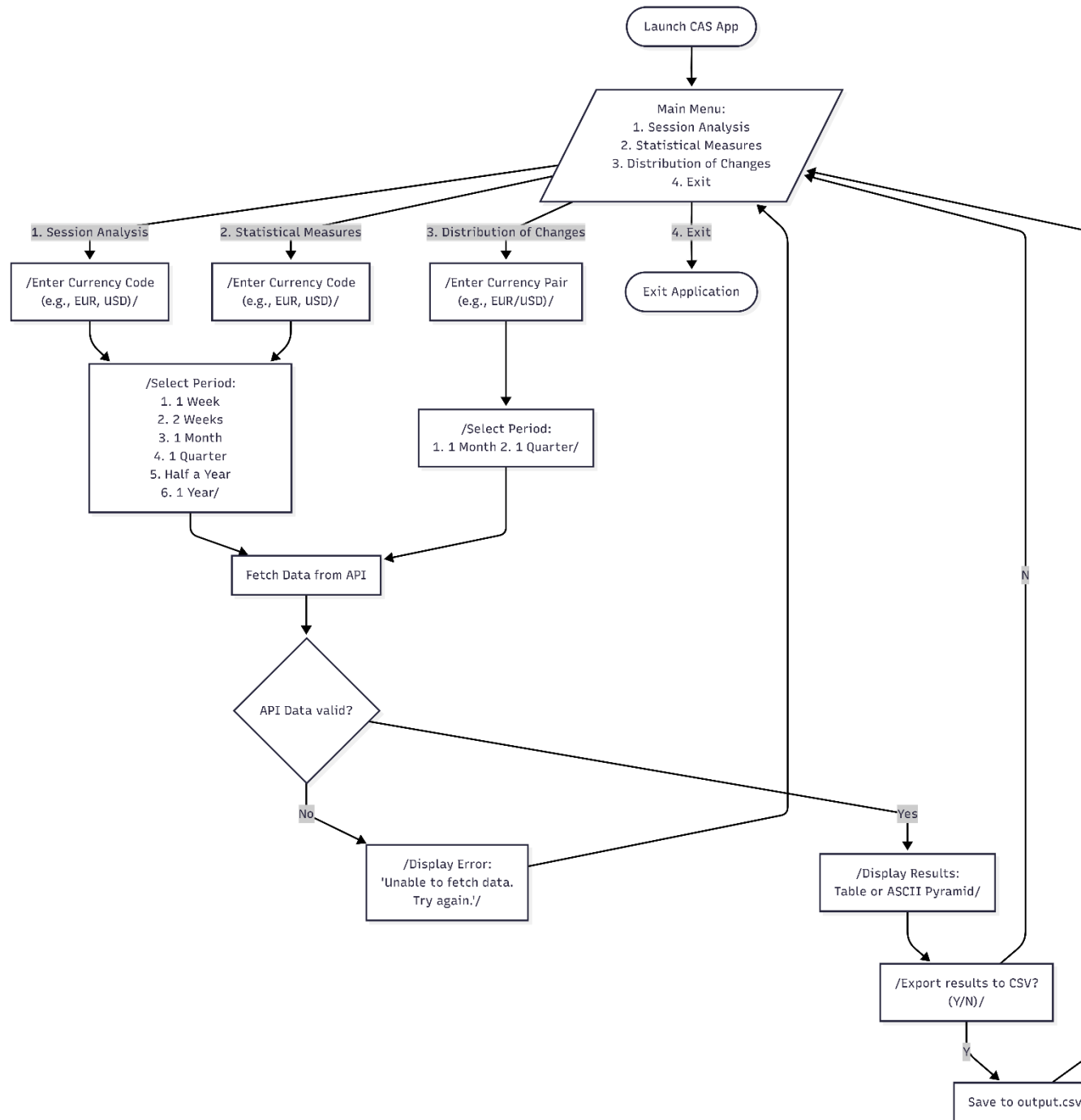


## 4. Use Case Diagram





# 5. CLI Communication Protocol



# 6. Gantt Chart (Project Schedule)

